Math 2320 - FE 1.3 Review

1. For the series, do they converge or diverge?

   (a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

   (b) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{\sqrt{n^5+4}}$

   (c) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{3^n}$

   (d) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$

   (e) $\displaystyle\sum_{n=1}^{\infty} \left[\frac{3n+5}{5n-1}\right]^n$

   (f) $\displaystyle\sum_{n=1}^{\infty} \frac{n^3}{n!}$

   (g) $\displaystyle\sum_{n=1}^{\infty} \frac{n^3}{\sqrt{n!}}$

   (h) $\displaystyle\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$

2. Graph the following equations given in polar form. Show the table of polar points as in class.

   (a) $r = 4\cos(\theta)$

   (b) $r = 6\sin(\theta)$

   (c) $r = 1 + \sin(\theta)$

   (d) $r = \sin(2\theta)$

3. Translate the following equations from rectangular to polar.

   (a) $y = x^2 + x$

   (b) $x = y^2 + y$

   (c) $x^2 + y^2 = 4$

   (d) $x^2 + y^2/16 = 1$

4. Translate the following equations from polar to rectangular.

   (a) $r = 4$

   (b) $r = 4\cos(\theta)$

   (c) $r = 6\sin(\theta)$

   (d) $r = \theta$

   (e) $r = \frac{\sin(\theta)}{\cos^2(\theta)}$

   (f) $r^2 - 2r\cos(\theta) = 0$